Support Vector Machines

Margins:

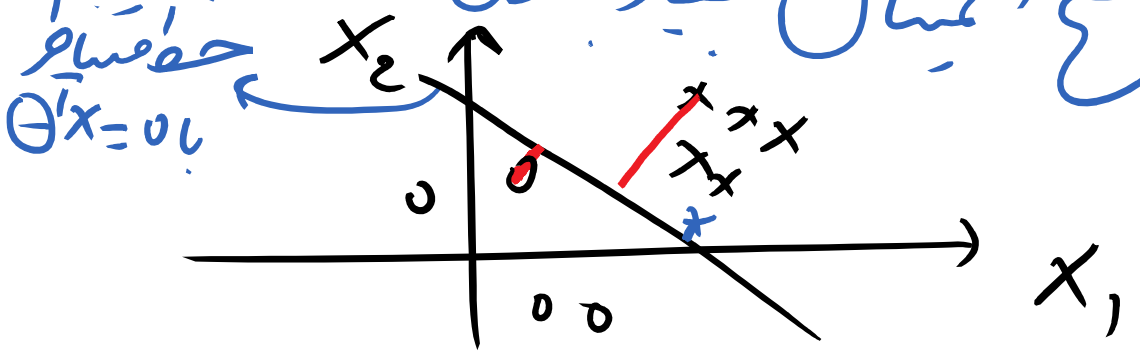
logReg $h_\theta(x) \geq 0.5 \equiv \theta^T x \geq 0 \Rightarrow y = 1$

$\theta^T x < 0 \Rightarrow y = 0$

اگر مدل به درستی کار کند f_{i+} برای $y^{(i)} = 1$ $\Leftarrow$ $\theta^T x^{(i)} \gg 0$

$(x^{(i)}, y^{(i)})$

$\Leftarrow$ سطح اطمینان بالایی دارد $\theta^T x^{(i)} \ll 0$ ، $y^{(i)} = 0$



Hyperplane : $w_0 + w_1 X_1 + w_2 X_2 + \dots w_p X_p = 0$

(فضای p بعدی)

$w_0 + w_1 X_1 = 0$ خط

$w_0 + w_1 X_1 + w_2 X_2 = 0$ صفحه

$w_0 + w_1 X_1 + \dots + w_p X_p > 0 \Rightarrow y = 1$

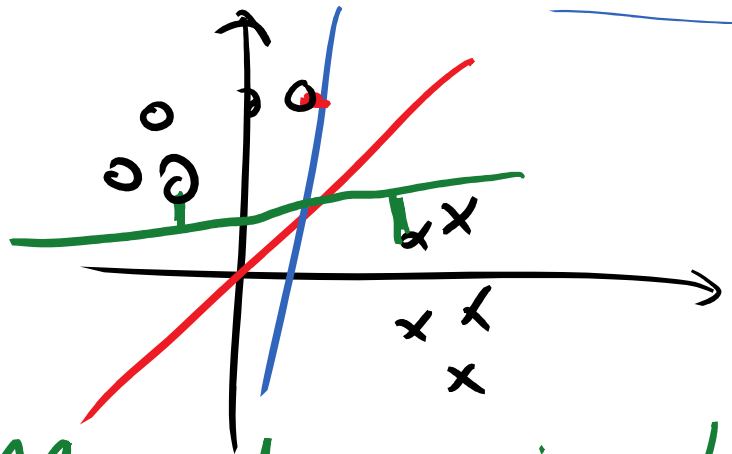
$< 0 \Rightarrow y = 0$

فضای p بعدی را به دو قسمت تقسیم می‌کند.

$y_i \in \{-1, 1\}$ Binary classification

$$y^{(i)}\left(w_1 x_1^{(i)} + w_2 x_2^{(i)} + \cdots w_n x_n^{(i)}\right) > 0$$

$i = 1 \ldots m$



$\{x^{(i)}, y^{(i)}\}$

$i = 1 \ldots m$

Maximal margin classifier:

- $X^{(i)}$ از هایپرپلین Margin کوتاه‌ترین فاصله عمودی مشاهده‌های

حاشیه:

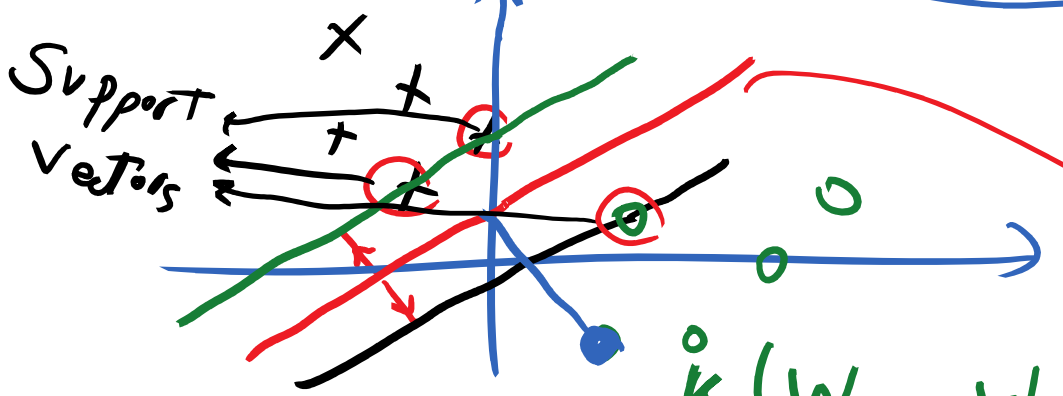
L M M classifier

$$\max_{W_0, W_1, \dots, W_n, M} M$$

subject to $\sum_{j=1}^{n} W_j^2 = 1$

$$y^{(i)}\left(W_0 + W_1 x_1^{(i)} + \dots + W_n x_n^{(i)}\right) \geq M$$

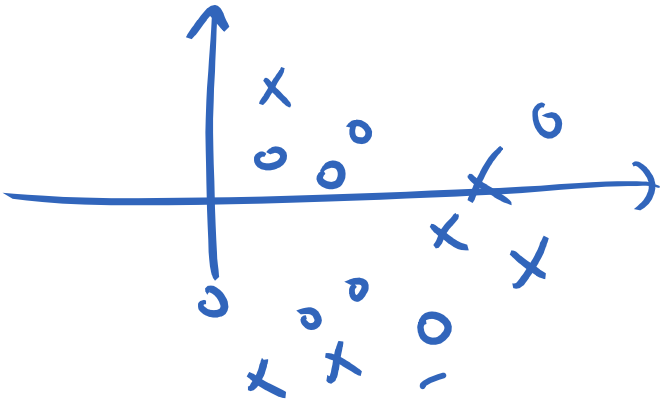
$\forall i = 1 \dots m$



MMH

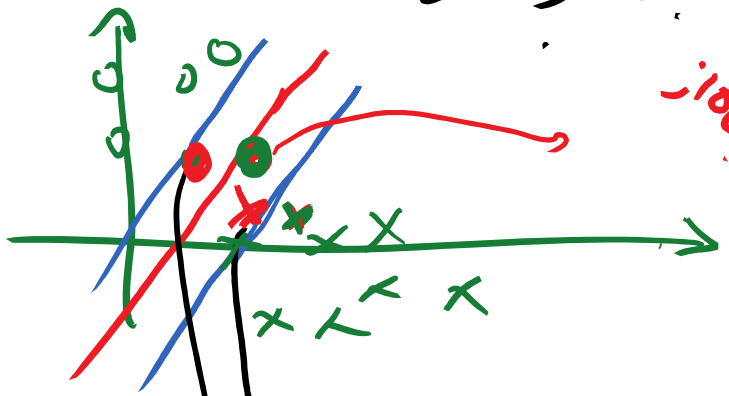
$$k(W_0 + W_1 X_1 + \dots + W_n X_n) = 0$$

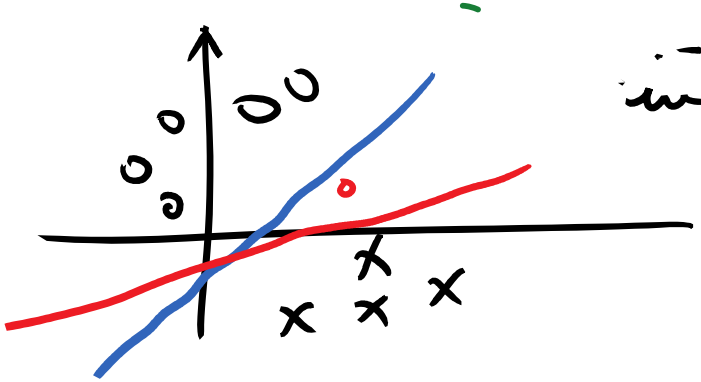
2_ Support vector classifier:



۱_ برای داده‌های جدید مقاوم است

۲_ جداسازی بهتری برای اکثر مشاهدات صورت می‌دهد.





margin حاشیه

بردار پشتیبان

hyperplane

Support vector classifier:

$$\max_{w_0, w_1, \dots, w_n, \epsilon_1, \epsilon_2, \dots, \epsilon_m, M} M$$

Subject to $\sum_{j=1}^{n} w_j^2 = 1$

$$y^{(i)}(w_0 + w_1 x_1^{(i)} + \dots + w_n x_n^{(i)}) \geq M(1 - \epsilon_i)$$

$$\epsilon_i \geq 0 \qquad \sum_{i=1}^{m} \epsilon_i \leq C \qquad C > 0$$

C: عددی دلخواه و مثبت

hyperplane, margin

MMC ← C = 0

- $x^{(i)}$ روی مرز margin است ← $\epsilon_i = 0$
- $x^{(i)}$ ... ← $\epsilon_i > 0$
- $x^{(i)}$ ... hyperplane ← $\epsilon_i > 1$

$C \uparrow \rightarrow$ Margin $\uparrow$

$C \uparrow \rightarrow$ bias $\uparrow$, var $\downarrow$

$C \downarrow \rightarrow$ bias $\downarrow$, var $\uparrow$

فقط نقاطی که روی حاشیه (مرز) قرار می‌گیرند یا از آن تجاوز می‌کنند تعیین کننده hyperplane هستند ← بردارهای پشتیبان (SV)

C ↑ ⇒ تعداد SV ↑ ، واریانس ↓

C ↓ ⇒ ↓ ، ، ، ، bias ↓

3- Support vector Machine

SV classifier + Kernel

$x \longrightarrow \phi(x)$

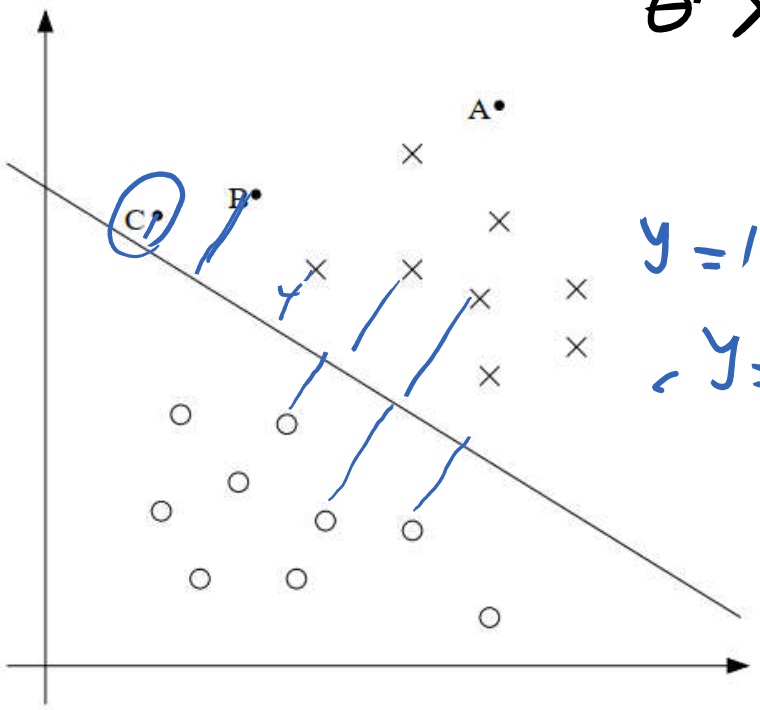
Margins

Logistic Regression $h_\theta = g(\theta^T x) \geq .5 \rightarrow y = 1$

$< .5 \rightarrow y = 0$

$\theta^T x \geq 0 \rightarrow y = 1$

$< 0 \rightarrow y = 0$



$y = 1 \quad \theta^T x \gg 0$

$y = 0 \quad \theta^T x \ll 0$

Hyperplane : $w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b$

$\theta^T X = \theta_n x_n + \theta_{n-1} x_{n-1} + \dots + \theta_1 x_1 + \theta_0 x_0 = 0$

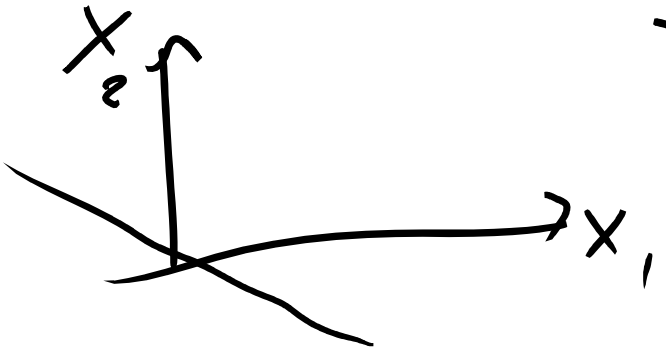
$\begin{bmatrix} 1 \\ x_1 \\ x_n \end{bmatrix}$

$\theta^T X = w^T X + b$

$\begin{bmatrix} x_1 \\ x_n \end{bmatrix}$

$w^T = [\theta_1 \dots \theta_n]$ | $b = \theta_0$

$x_2 + 2 x_1 = 0$



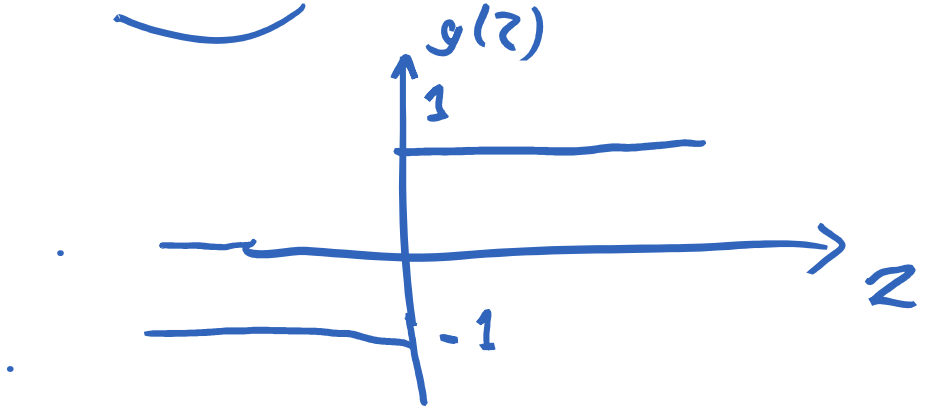
Notation

- use $y \in \{-1,1\}$ (instead of $\{0,1\}$) to denote the class labels.

- we will use parameters w, b, and write our classifier as:

$$h_{w,b}(x) = g\left(w^T x + b\right).$$

Here, $g(z) = 1$ if $z \geq 0$, and $g(z) = -1$ otherwise.



Functional and geometric margins

Given a training example $(x^{(i)}, y^{(i)})$

we define the functional margin of (w, b) with respect to the training example as

functional margin

$$\gamma^{(i)} = y^{(i)}(w^T x^{(i)} + b)$$

for the functional margin to be large, we need:

if $y^{(i)} = 1$:

we need $w^T x^{(i)} + b$ to be a large positive number

if $y^{(i)} = -1$:

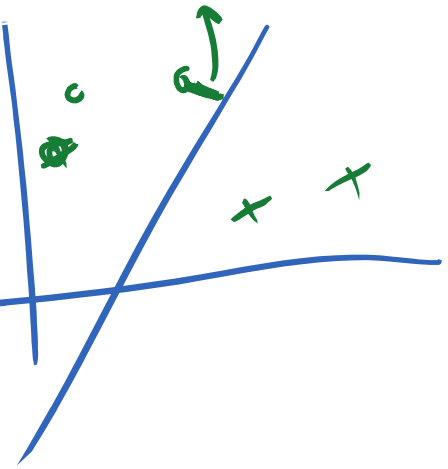
we need $w^T x^{(i)} + b$ to be a large negative number

Hence, a large functional margin represents a confident and a correct prediction.

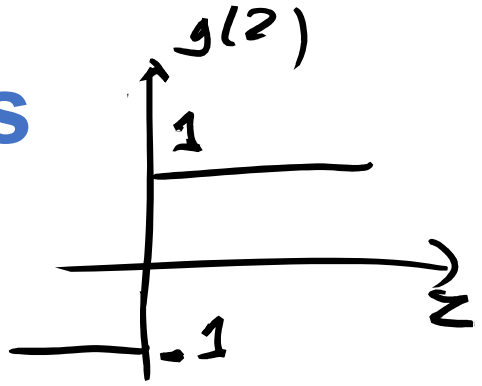
Functional and geometric margins

Given a training set $S = \{(x^{(i)}, y^{(i)}); i = 1, \ldots, m\}$, the functional margin of the hyperplane (w,b) w.r.t S denoted by $\hat{\gamma}$, can be written:

$$\hat{\gamma} = \min_{i=1,\ldots m} \hat{\gamma}^{(i)}.$$



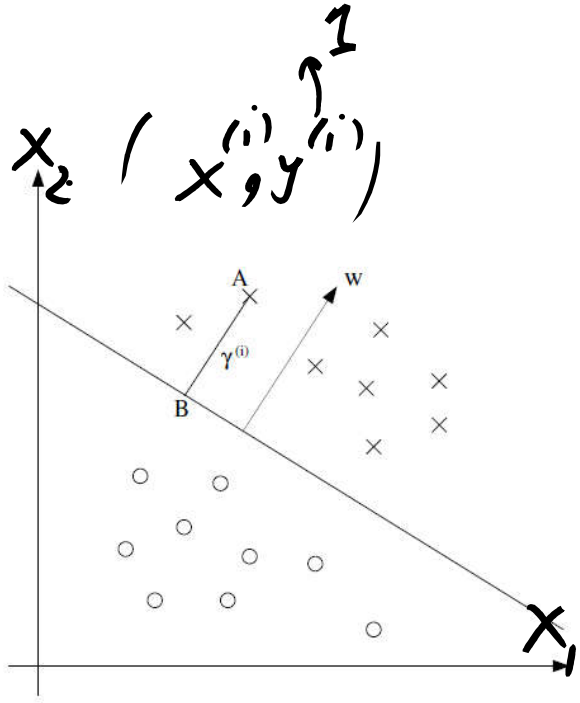
Functional and geometric margins



- if we replace w with $2w$ and b with $2b$

$$g(w^T x + b) = g(2w^T x + 2b)$$

- this would not change $h_{w,b}(x)$ at all. and hence also $h_{w,b}(x)$, depends only on the sign, but not on the magnitude, of $w^T x + b$.
- But results in multiplying our functional margin by a factor of 2.
- So we use geometric margin $\gamma^{(i)}$



Functional and geometric margins

How can we find the value of $\gamma^{(i)}$?

Well, $w/\| w \|$ is a unit-length vector pointing in the same direction as w.

Since A represents $x^{(i)}$, we therefore find that the point B is given by

$$x^{(i)} - \gamma^{(i)} \cdot w/\| w \|.$$

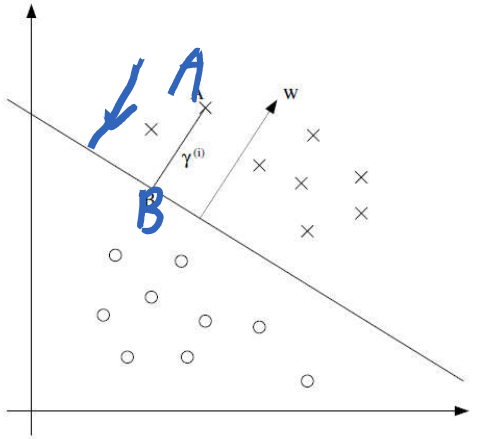
But this point lies on the decision boundary, and all points x on the decision boundary satisfy the equation $w^T x + b = 0$. Hence:

$$w^T \left(x^{(i)} - \gamma^{(i)} \frac{w}{\| w \|} \right) + b = 0.$$

Solving for $\gamma^{(i)}$ yields:

$$\gamma^{(i)} = \frac{w^T x^{(i)} + b}{\| w \|} = \left(\frac{w}{\| w \|} \right)^T x^{(i)} + \frac{b}{\| w \|}.$$

Geometric margin



Functional and geometric margins

we define the geometric margin of (w, b) with respect to a training example $\left(x^{(i)}, y^{(i)}\right)$ to be:

$$\gamma^{(i)} = y^{(i)} \left(\left(\frac{w}{\| w \|} \right)^T x^{(i)} + \frac{b}{\| w \|} \right)$$

Given a training set $S = \left\{ \left(x^{(i)}, y^{(i)}\right); i = 1, \ldots, m \right\}$, the geometric margin of the hyperplane (w,b) w.r.t S denoted by γ, can be written:

$$\gamma = \min_{i=1,\ldots m} \gamma^{(i)}$$

The optimal margin classier

MMC

- we will assume that we are given a training set that is linearly separable
- How will we find the one that achieves the **maximum geometric margin**? We can pose the following optimization problem:

$$\max_{\gamma,w,b} \quad \gamma$$
$$\text{s.t.} \quad y^{(i)}\left(w^T x^{(i)} + b\right) \geq \gamma,\ i = 1, \ldots, m$$
$$\| w \| = 1$$

  - we want to maximize γ, subject to each training example having functional margin at least γ.

The optimal margin classier

- If we could solve the optimization problem above, we'd be done. But the "|w|=1" constraint is a nasty (non-convex) one, and this problem certainly isn't in any format that we can plug into standard optimization software to solve.
- So, let's try transforming the problem:

functional margin

$$\max_{\hat{\gamma},w,b} \quad \frac{\hat{\gamma}}{\| w \|}$$

$$\text{s.t.} \quad y^{(i)}\left(w^T x^{(i)} + b\right) \geq \hat{\gamma}, \; i = 1, \ldots, m$$

  - Here, we're going to maximize $\hat{\gamma}/\| w \|$, subject to the functional margins all being at least $\hat{\gamma}$.
  - we've gotten rid of the constraint $\| w \| = 1$ that we didn't like. The downside is that we now have a nasty (again, non-convex) objective $\frac{\hat{\gamma}}{\|w\|}$ function.

The optimal margin classier

$w^T x + b = 0$
$2w^T x + 2b = 0$

- Recall our earlier discussion that we can add an arbitrary scaling constraint on w and b without changing anything.

- We will introduce the scaling constraint that the functional margin of w, b with respect to the training set must be 1 : $\hat{\gamma} = 1.$

- Plugging this into our problem above, and noting that maximizing $\hat{\gamma}/\| w \| = 1/\| w \|$ is the same thing as minimizing $\| w \|^2$, we now have the following optimization problem:

$$\min_{w,b} \quad \frac{1}{2} \| w \|^2$$
$$\text{s.t.} \quad y^{(i)}\left(w^T x^{(i)} + b\right) \geq 1, \; i = 1, \ldots, m$$

The above is an optimization problem with a **convex quadratic objective** and only *linear constraints*.

The optimal margin classier

$$\min_{w,b} \quad \frac{1}{2} \| w \|^2$$
$$\text{s.t.} \quad y^{(i)}\left(w^T x^{(i)} + b\right) \geq 1, \; i = 1, \ldots, m$$

- Its solution gives us the **optimal margin classifier**. This optimization problem can be solved using commercial quadratic programming (QP).
- What we will still do is to make a digression to talk about **Lagrange duality**.
- This will lead us to our optimization problem's dual form, which will play a key role in allowing us to use **kernels** to get optimal margin classifiers to work efficiently in very high dimensional spaces.
- The dual form will also allow us to derive an efficient algorithm for solving the above optimization problem that will typically do much better than generic QP software.

Lagrange duality

- Consider a problem of the following form:

$$\begin{aligned} \min_{w} \quad & f(w) \\ \text{s.t.} \quad & h_i(w) = 0,\ i = 1, \ldots, l \end{aligned}$$

- define the Lagrangian to be:

$$\mathcal{L}(w, \beta) = f(w) + \sum_{i=1}^{l} \beta_i h_i(w)$$

- Here, the β_i 's are called the Lagrange multipliers. We would then find and set $\mathcal{L}$'s partial derivatives to zero:

$$\frac{\partial \mathcal{L}}{\partial w_i} = 0; \quad \frac{\partial \mathcal{L}}{\partial \beta_i} = 0,$$

and solve for w and β.

Lagrange duality

Consider the following, which we'll call the primal optimization problem:

$$\begin{aligned} \min_{w} \quad & f(w) \\ \text{s.t.} \quad & g_i(w) \leq 0, \; i = 1, \ldots, k \\ & h_i(w) = 0, \; i = 1, \ldots, l. \end{aligned}$$

To solve it, we start by defining the generalized Lagrangian

$$\mathcal{L}(w, \alpha, \beta) = f(w) + \sum_{i=1}^{k} \alpha_i g_i(w) + \sum_{i=1}^{l} \beta_i h_i(w)$$

Here, the α_i 's and β_i 's are the Lagrange multipliers.

Lagrange duality

Consider the quantity

$$\theta_{\mathcal{P}}(w) = \max_{\alpha,\beta:\alpha_i \geq 0} \mathcal{L}(w, \alpha, \beta).$$

Here, the " $\mathcal{P}$ " subscript stands for "primal." Let some w be given. If w violates any of the primal constraints (i.e., if either $g_i(w) > 0$ or $h_i(w) \neq 0$ for some i, then you should be able to verify that

$$\theta_{\mathcal{P}}(w) = \begin{cases} f(w) & \text{if } w \text{ satisfies primal constraints} \\ \infty & \text{otherwise.} \end{cases}$$

Lagrange duality

$$\min_{w} \theta_{\mathcal{P}}(w) = \min_{w} \max_{\alpha,\beta : \alpha_i \geq 0} \mathcal{L}(w, \alpha, \beta),$$

مسئله اصلی

$$d^* = \max_{\alpha,\beta : \alpha_i \geq 0} \min_{w} \mathcal{L}(w, \alpha, \beta) \leq \min_{w} \max_{\alpha,\beta : \alpha_i \geq 0} \mathcal{L}(w, \alpha, \beta) = p^*.$$

حل مسئله dual

مسئله dual

$d^* = p^*$ ← (KKT) شرایط تحت و Convex باشد مسئله اگر

Optimal margin classifiers

$$\min_{w,b} \quad \frac{1}{2} \| w \|^2$$
$$\text{s.t.} \quad y^{(i)}\left(w^T x^{(i)} + b\right) \geq 1, \; i = 1, \ldots, m$$

When we construct the Lagrangian for our optimization problem we have:

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} \| w \|^2 - \sum_{i=1}^{m} \alpha_i \left[y^{(i)}\left(w^T x^{(i)} + b\right) - 1 \right] \qquad (21)$$

Let's find the dual form of the problem. which we'll do by setting the derivatives of $\mathcal{L}$ with respect to w and b to zero. We have:

$$\nabla_w \mathcal{L}(w, b, \alpha) = w - \sum_{i=1}^{m} \alpha_i y^{(i)} x^{(i)} = 0$$

$$w = \sum_{i=1}^{m} \alpha_i y^{(i)} x^{(i)} \qquad (22)$$

$$\frac{\partial}{\partial b} \mathcal{L}(w, b, \alpha) = \sum_{i=1}^{m} \alpha_i y^{(i)} = 0 \qquad (23)$$

Optimal margin classifiers

If we take the definition of w in Equation (22) and plug that back into the Lagrangian (Equation 21), and simplify, we get

$$\mathcal{L}(w, b, \alpha) = \sum_{i=1}^{m} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{m} y^{(i)} y^{(j)} \alpha_i \alpha_j \left(x^{(i)}\right)^T x^{(j)} - b \sum_{i=1}^{m} \alpha_i y^{(i)}$$

But from Equation (23), the last term must be zero, so we obtain

$$\mathcal{L}(w, b, \alpha) = \sum_{i=1}^{m} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{m} y^{(i)} y^{(j)} \alpha_i \alpha_j \left(x^{(i)}\right)^T x^{(j)}$$

$\langle x^{(i)}, x^{(j)} \rangle$

Optimal margin classifiers

Recall that we got to the equation above by minimizing $\mathcal{L}$ with respect to w and b. Putting this together with the constraints $\alpha_i \geq 0$ (that we always had) and the constraint (23), we obtain the following **dual optimization problem**:

$$\max_{\alpha} \quad W(\alpha) = \sum_{i=1}^{m} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{m} y^{(i)} y^{(j)} \alpha_i \alpha_j \langle x^{(i)}, x^{(j)} \rangle$$

$$\text{s.t.} \quad \alpha_i \geq 0, \; i = 1, \ldots, m$$

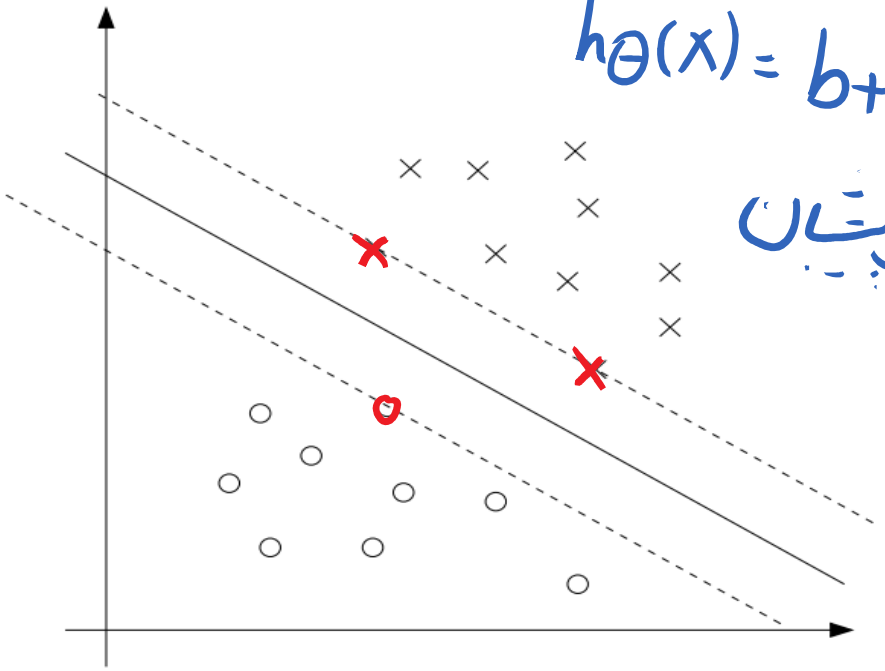
$$\sum_{i=1}^{m} \alpha_i y^{(i)} = 0$$

Optimal margin classifiers

$$w^* = \sum_{i=1}^{m} \alpha_i y^{(i)} x^{(i)}$$

$$b^* = -\frac{\max_{i:y^{(i)}=-1} w^{*T} x^{(i)} + \min_{i:y^{(i)}=1} w^{*T} x^{(i)}}{2}.$$

Optimal margin classifiers



here, these are the three points (one negative and two positive examples)
Thus, only three of the α_i 's - namely, the ones corresponding to these three training examples - will be non-zero at the optimal solution to our optimization problem.
These three points are called the **support vectors** in this problem.

Regularization and the non-separable case

- mapping data to a high dimensional feature space via ϕ does generally increase the likelihood that the data is separable, we can't guarantee that it always will be so.

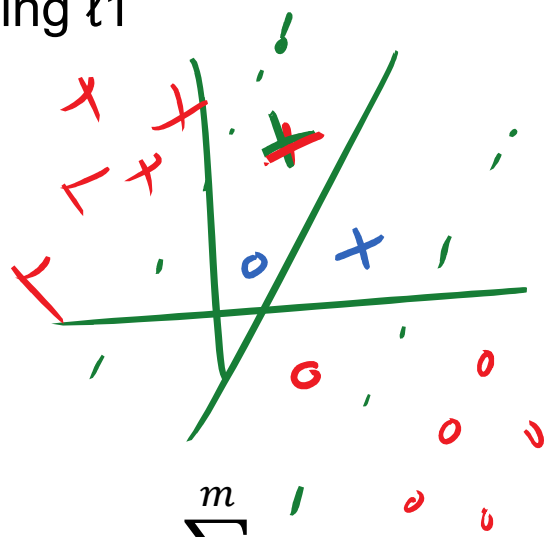
support vector classifier

- To make the algorithm work for non-linearly separable datasets as well as be less sensitive to outliers, we reformulate our optimization (using ℓ1 regularization) as follows:

$$\min_{\gamma,w,b} \quad \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{m}\xi_i$$

$$\text{s.t.} \quad y^{(i)}\left(w^T x^{(i)} + b\right) \geq 1 - \xi_i, \; i = 1, \ldots, m$$

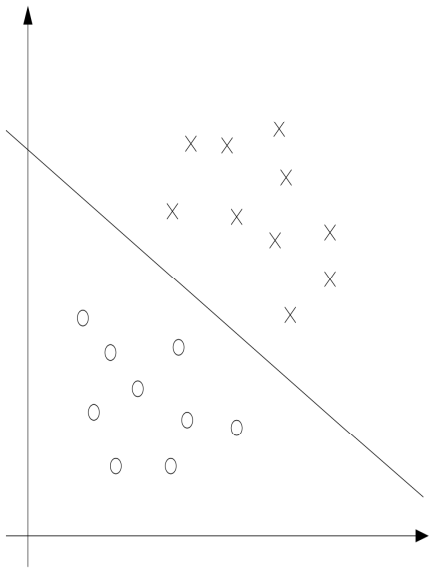
$$\xi_i \geq 0, \; i = 1, \ldots, m$$

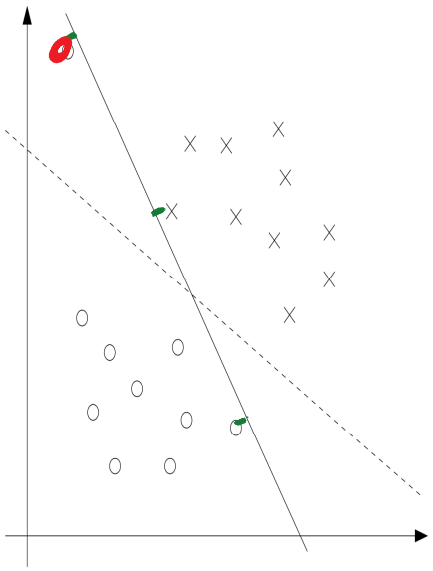


As before, we can form the Lagrangian:

$$\mathcal{L}(w, b, \xi, \alpha, r) = \frac{1}{2}w^T w + C\sum_{i=1}^{m}\xi_i - \sum_{i=1}^{m}\alpha_i\left[y^{(i)}\left(x^T w + b\right) - 1 + \xi_i\right] - \sum_{i=1}^{m} r_i\xi_i$$

Regularization and the non-separable case





Regularization and the non-separable case

Thus, examples are now permitted to have (functional) margin less than 1, and if an example has functional margin $1 - \xi_i$ (with $\xi > 0$), we would pay a cost of the objective function being increased by $C\xi_i$. The parameter C controls the relative weighting between the twin goals of making the $||w||^2$ small (which we saw earlier makes the margin large) and of ensuring that most examples have functional margin at least 1.

Support vector classifier:

$$\max_{w_0, w_1, \dots, w_n, \epsilon_1, \epsilon_2, \dots, \epsilon_m, M} M$$

Subject to

$$\sum_{j=1}^{n} w_j^2 = 1$$

$$y^{(i)}\left(w_0 + w_1 x_1^{(i)} + \dots + w_n x_n^{(i)}\right) \geq M(1 - \epsilon_i)$$

$$\epsilon_i \geq 0 \qquad \sum_{i=1}^{m} \epsilon_i \leq C \qquad C > 0$$

C: هایپرپارامتر

hyperplane, margin

MMC ← C=0

$C \uparrow \rightarrow$ Margin ↑

$C \uparrow \rightarrow$ bias ↑, var ↓

$C \downarrow \rightarrow$ bias ↓, var ↑

$x^{(i)}$ در سمت درست margin است ← $\epsilon_i = 0$

$x^{(i)}$ روی margin است ← $1 > \epsilon_i > 0$

$x^{(i)}$ روی hyperplane است ← $\epsilon_i > 1$

فقط نقاطی که روی حاشیه (مرز) قرار دارند یا از آن تجاوز کرده عبور کرده هستند ← بردارهای پشتیبان (SV) hyperplane

C↑ ⇒ تعداد SV ↑ ، var ↓

C↓ ⇒ ↓ ، bias ↓

overfit
underfit

var↑
bias↑

3- Support vector Machine

SV classifier + Kernel

$X \longrightarrow \phi(x)$

Regularization and the non-separable case

dual form of the problem:

$$\max_{\alpha} \quad W(\alpha) = \sum_{i=1}^{m} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{m} y^{(i)} y^{(j)} \alpha_i \alpha_j \langle x^{(i)}, x^{(j)} \rangle$$

$$\text{s.t.} \quad 0 \le \alpha_i \le C, \; i = 1, \ldots, m$$

$$\sum_{i=1}^{m} \alpha_i y^{(i)} = 0$$

$$w = \sum_{i=1}^{m} \alpha_i y^{(i)} x^{(i)}$$

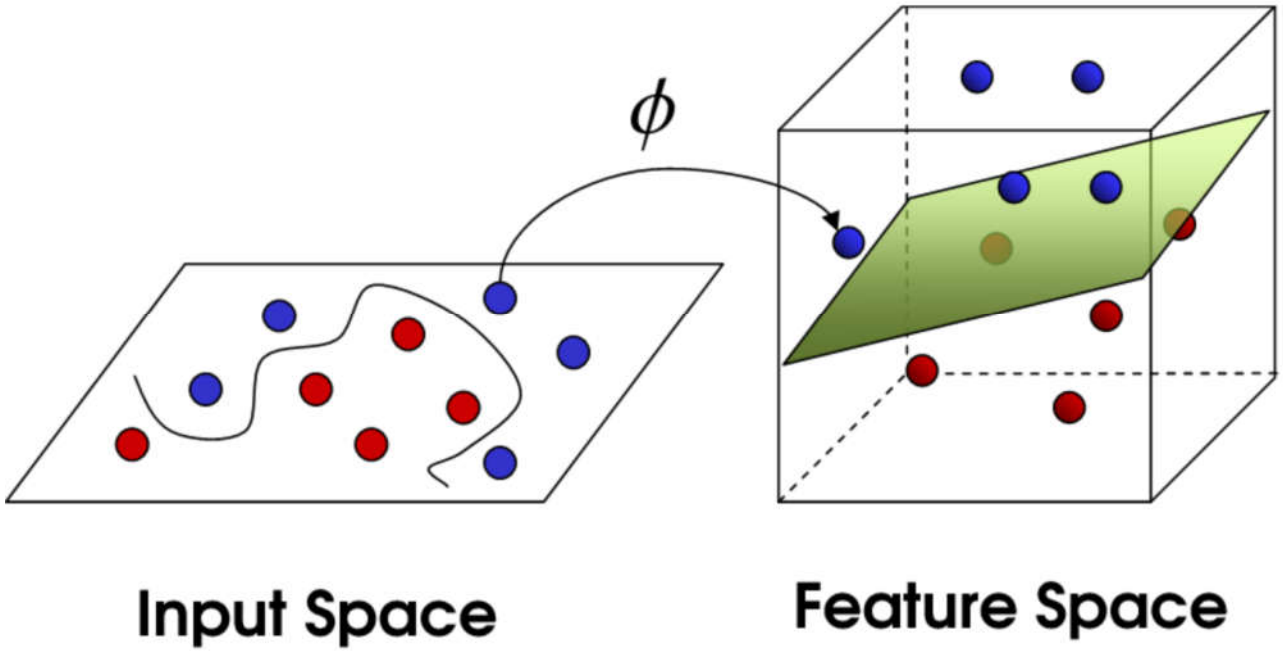
$$h_\theta(x) = w^T x + b = \sum_{i \in S} \alpha_i y^{(i)} \langle x, x^{(i)} \rangle + b \gtrless 0$$

$$\langle x, x^{(i)} \rangle \rightarrow k(x, x^{(i)})$$

(SVM)

SVM (Non-linear decision boundary)

$X \rightarrow \phi(X)$

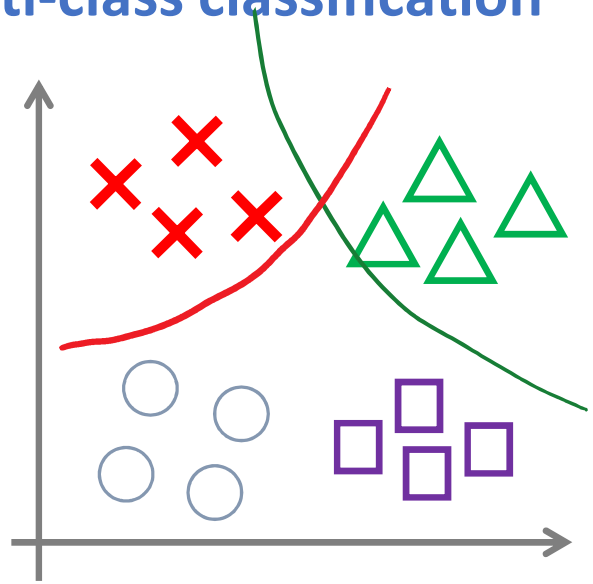


SVM (Non-linear decision boundary)

Common kernel functions include:

- Linear: $\langle x, x' \rangle$
- Polynomial: $(\gamma \langle x, x' \rangle + r)^d$
- rbf: $\exp(-\gamma |x - x'|^2)$ where γ must be greater than 0.
- Sigmoid $(\tanh(\gamma \langle x, x' \rangle + r))$

Multi-class classification



$$y \in \{1, 2, 3, \ldots, K\}$$

$$\max_{k} \; h_\theta^{(k)}(x_{test}) = y_{test}$$

$$k = 1 \ldots K$$

Many SVM packages already have built-in multi-class classification functionality.

Otherwise, use one-vs.-all method. (Train K SVMs, one to distinguish $y = i$ from the rest, for $i = 1, 2, \ldots, K$), get $\theta^{(1)}, \theta^{(2)}, \ldots, \theta^{(K)}$

Pick class i with largest $(\theta^{(i)})^T x$

Logistic regression vs. SVMs

n =number of features ($x \in \mathbb{R}^{n+1}$), m =number of training examples

- If n is large (relative to m):
  Use logistic regression, or SVM without a kernel (“linear kernel”)

  If n is small, m is intermediate:
  Use SVM with Gaussian kernel

- If n is small, m is large:
  Create/add more features, then use logistic regression or SVM without a kernel

Neural network is likely to work well for most of these settings, but may be slower to train.